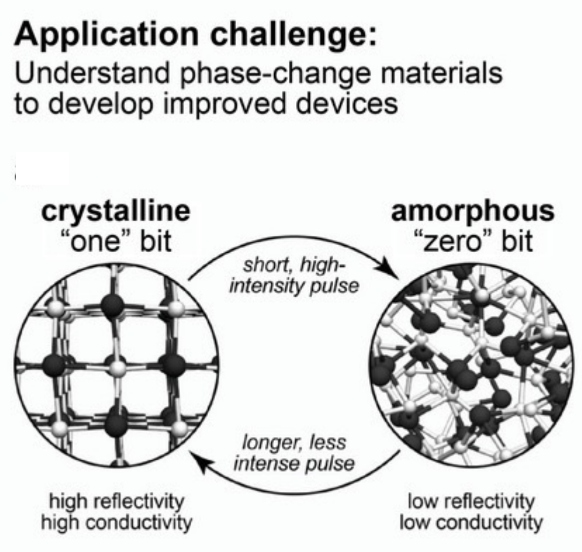
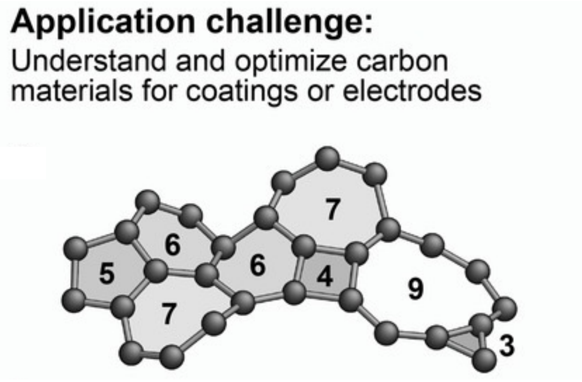
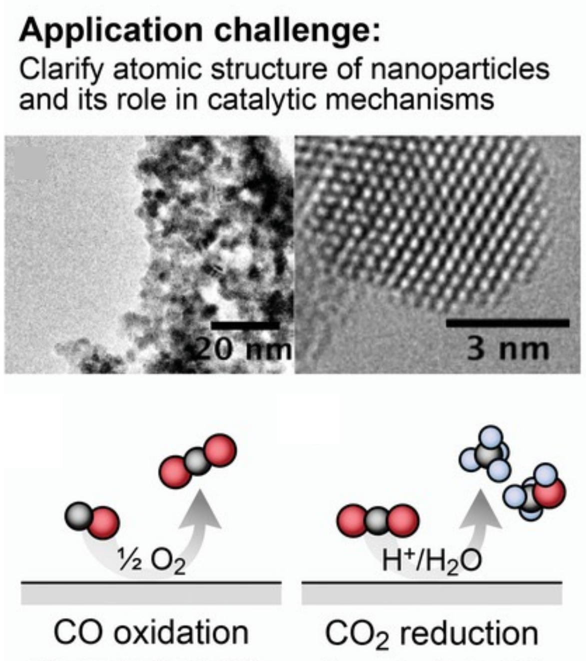
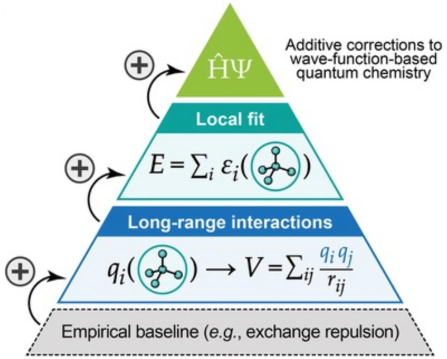
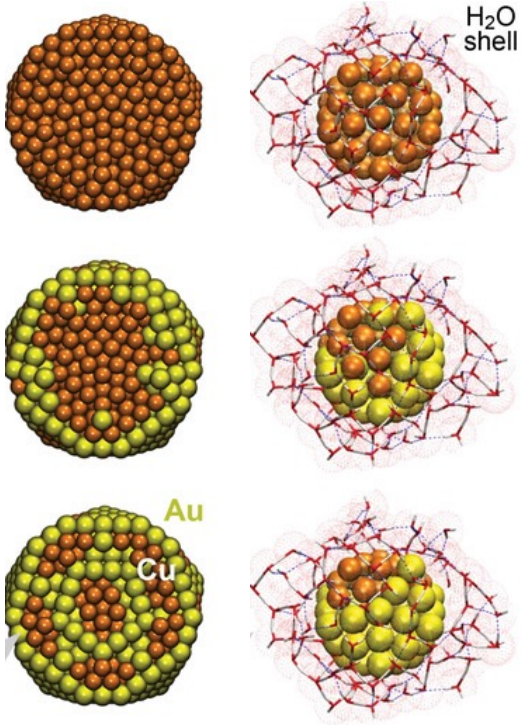
Interatomic potentials: use cases

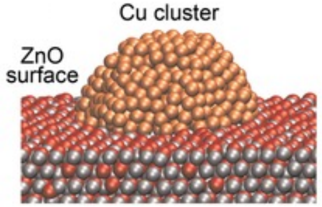


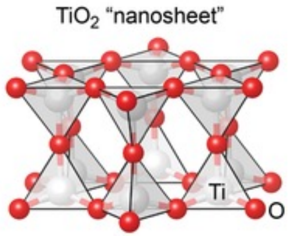




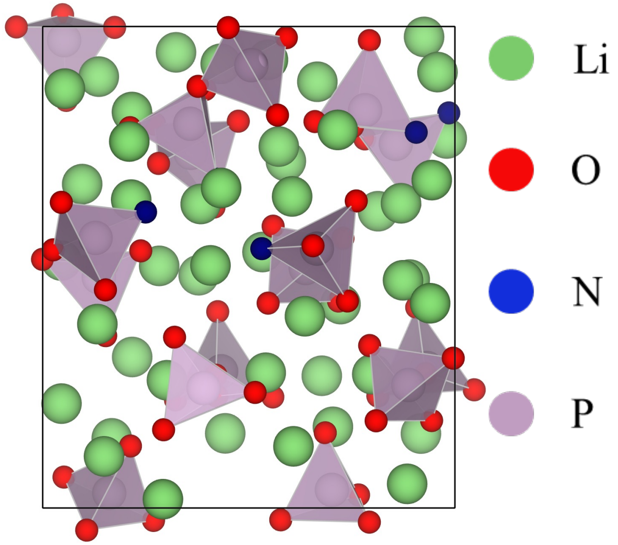








Modelling amorphous electrolytes and interfaces

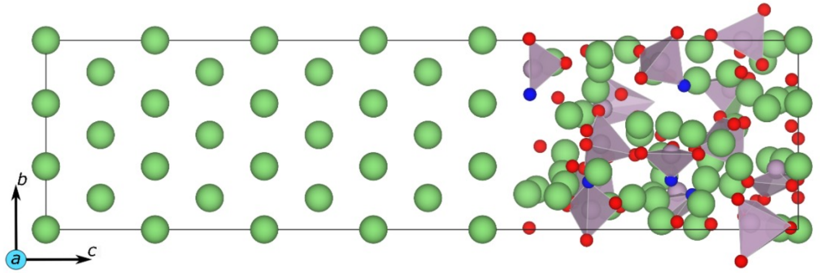


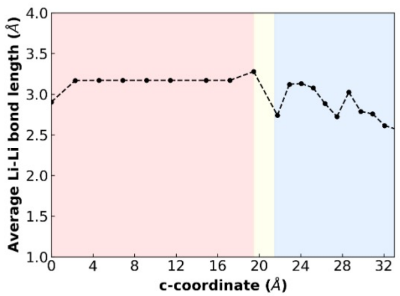
Li-P-O-N

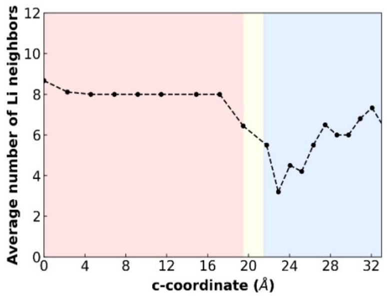
Amorphous solid electrolytes
- Reasonable Li-conductivity, hinder dendrite growth
- Case of LiPON

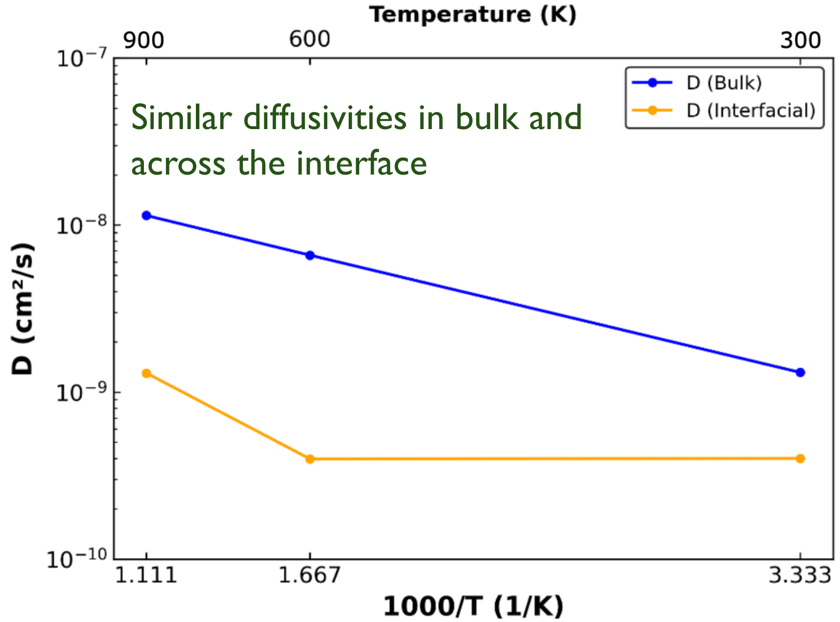
Modelling amorphous structures
- Non-trivial with classical *ab initio*
- Employ machine learned interatomic potentials (MLIPs)

Analyze local structure and Li-ion conductivity with MLIPs
- Training dataset: ~13000 structures





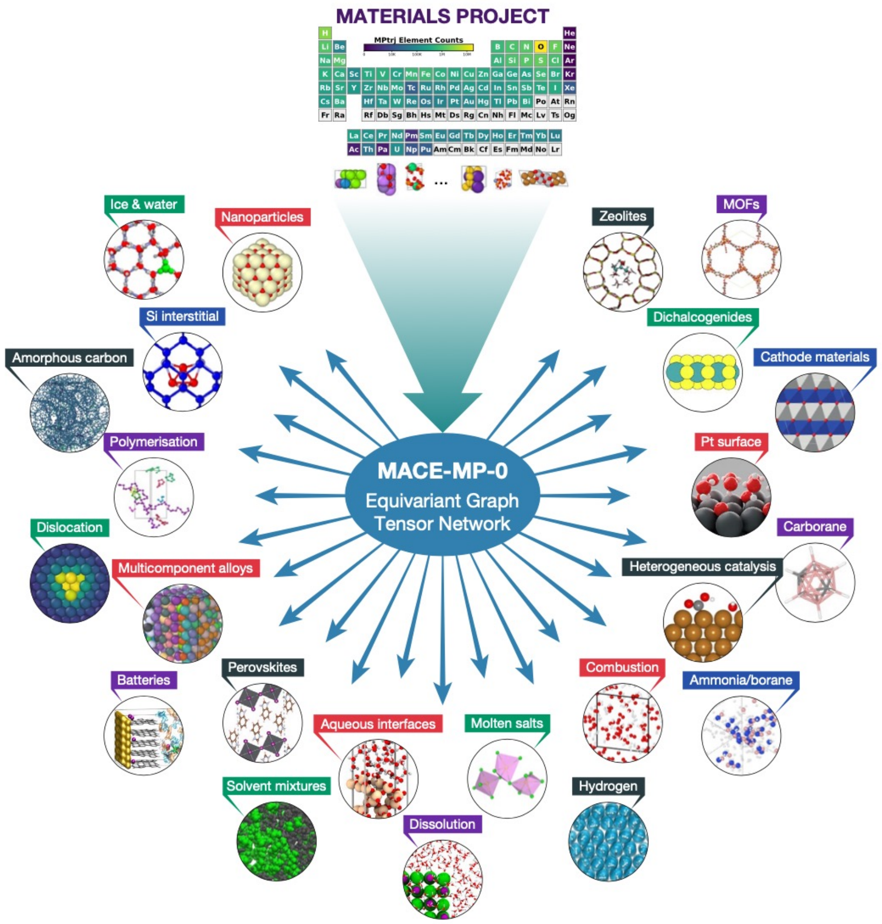




MLIP: Neural Equivariant Interatomic Potential (NequIP)

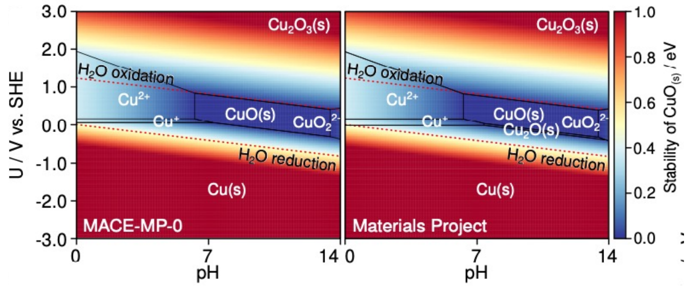
A. Seth, R.P. Kulkarni and G. Sai Gautam, **ACS Mater. Au 2025**, 5, 458-468

Foundational interatomic potentials



Trained on Materials Project trajectory dataset (~1.5M structures)

- Stable performance on 30 different property predictions/application areas
- Stable dynamics in solids, liquids, and gases
- GPU; limited system size



Many more to come…

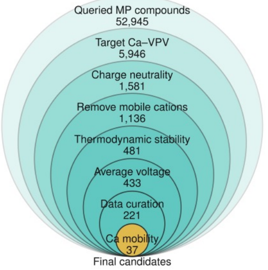
Model	F1 ↑	DAF ↑	Prec ↑	Acc ↑	MAE ↓	R^2 ↑	κ_{SRME} ↓	Training Set
eqV2 S DeNS	0.815	5.042	0.771	0.941	0.036	0.788	1.665	146k (1.58M) (MPtrj)
ORB MPtrj	0.765	4.702	0.719	0.922	0.045	0.756	1.725	146k (1.58M) (MPtrj)
SevenNet-l3i5	0.76	4.629	0.708	0.92	0.044	0.776	0.55	146k (1.58M) (MPtrj)
SevenNet-0	0.724	4.252	0.65	0.904	0.048	0.75	0.767	146k (1.58M) (MPtrj)
GRACE-2L (r6)	0.691	4.163	0.636	0.896	0.052	0.741	0.525	146k (1.58M) (MPtrj)
MACE-MP-0	0.669	3.777	0.577	0.878	0.057	0.697	0.647	146k (1.58M) (MPtrj)
CHGNet	0.613	3.361	0.514	0.851	0.063	0.689	1.717	146k (1.58M) (MPtrj)
M3GNet	0.569	2.882	0.441	0.813	0.075	0.585	1.412	62.8k (188k) (MPF)

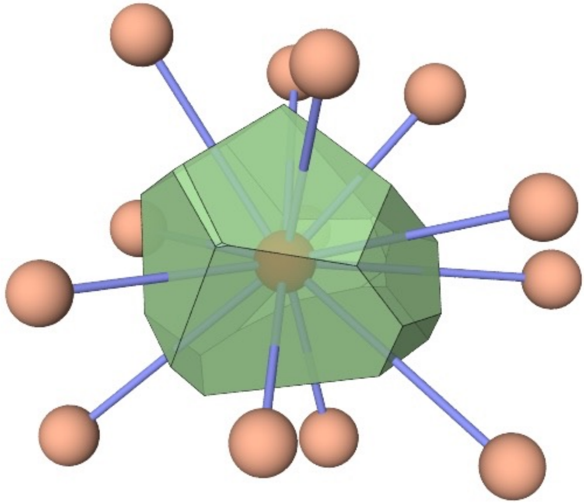
https://matbench-discovery.materialsproject.org/

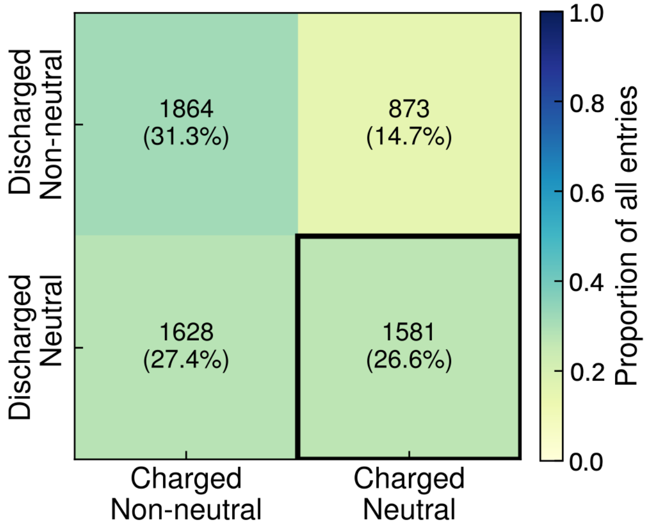
Batatia et al., arXiv, 2401.00096v2 (2024)

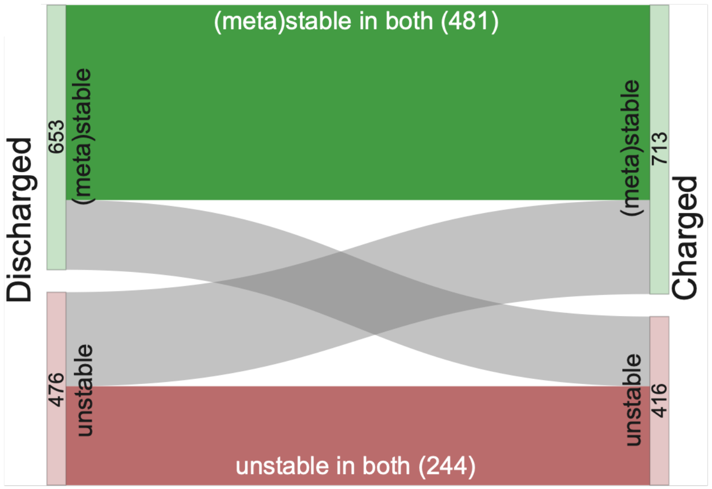
Using interatomic potentials: accelerated discovery of Ca-battery electrodes

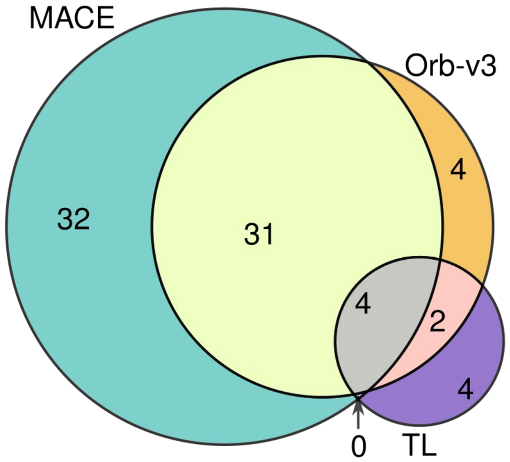
- Ca-batteries: promising alternative battery technology, but requires positive electrodes (cathodes)
  - Geometry-driven, ML-accelerated discovery of 37 promising candidates











D.B. Tekliye, A.K. Bheemaguli, and G. Sai Gautam, **arXiv 2026**